

$$= 0.60 \times 0.20 + 0.25 \times 0.80 = 0.32.$$

Now we can implement the Bayes' formula answer to our question:

$$P(\text{tech} | R > 10\%) = \frac{P(R > 10\% | \text{tech}) \times P(\text{tech})}{P(R > 10\%)} = \frac{0.60 \times 0.20}{0.32} = 0.375.$$

The probability that a firm with a return of >10 percent is a tech firm is 0.375, which is impressive because the probability that a firm is a tech firm (from the whole sample) is only 0.20. In sum, it can be readily seen from the tree map and the underlying frequency data (Exhibit 4 and 5, respectively) or from the probabilities in Bayes' formula that 160 firms have $R > 10$ percent, and 60 of them are tech firms, so

$$P(\text{tech} | R > 10\%) = 60/160 = 0.375.$$

Users of Bayesian statistics do not consider probabilities (or likelihoods) to be known with certainty but believe that these should be subject to modification whenever new information becomes available. Our beliefs or probabilities are continually updated as new information arrives over time.

To further illustrate Bayes' formula, we work through an investment example that can be adapted to any actual problem. Suppose you are an investor in the stock of DriveMed, Inc. Positive earnings surprises relative to consensus EPS estimates often result in positive stock returns, and negative surprises often have the opposite effect. DriveMed is preparing to release last quarter's EPS result, and you are interested in which of these three events happened: *last quarter's EPS exceeded the consensus EPS estimate*, *last quarter's EPS exactly met the consensus EPS estimate*, or *last quarter's EPS fell short of the consensus EPS estimate*. This list of the alternatives is mutually exclusive and exhaustive.

On the basis of your own research, you write down the following **prior probabilities** (or priors, for short) concerning these three events:

- $P(\text{EPS exceeded consensus}) = 0.45$
- $P(\text{EPS met consensus}) = 0.30$
- $P(\text{EPS fell short of consensus}) = 0.25$

These probabilities are "prior" in the sense that they reflect only what you know now, before the arrival of any new information.

The next day, DriveMed announces that it is expanding factory capacity in Singapore and Ireland to meet increased sales demand. You assess this new information. The decision to expand capacity relates not only to current demand but probably also to the prior quarter's sales demand. You know that sales demand is positively related to EPS. So now it appears more likely that last quarter's EPS will exceed the consensus.

The question you have is, "In light of the new information, what is the updated probability that the prior quarter's EPS exceeded the consensus estimate?"

Bayes' formula provides a rational method for accomplishing this updating. We can abbreviate the new information as *DriveMed expands*. The first step in applying Bayes' formula is to calculate the probability of the new information (here: *DriveMed expands*), given a list of events or scenarios that may have generated it. The list of events should cover all possibilities, as it does here. Formulating these conditional probabilities is the key step in the updating process. Suppose your view, based on research of DriveMed and its industry, is

$$P(\text{DriveMed expands} | \text{EPS exceeded consensus}) = 0.75$$

$$P(\text{DriveMed expands} | \text{EPS met consensus}) = 0.20$$

$$P(\text{DriveMed expands} | \text{EPS fell short of consensus}) = 0.05$$

Conditional probabilities of an observation (here: *DriveMed expands*) are sometimes referred to as **likelihoods**. Again, likelihoods are required for updating the probability.